Tenet: Agent Memory as a Self-Consistent World Model

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Abstract

Long-term memory for LLM agents is almost universally implemented as *retrieval over a growing log of past turns*—a document-retrieval abstraction. We argue this is the wrong abstraction for an agent that must model a changing world. We introduce **knowledge churn**—the repeated updating of a fact over a long interaction—and show that retrieval-augmented memory (RAG-memory) *silently degrades* under it: as the number of stale versions of a fact exceeds the retrieval budget k, the reader is handed conflicting values and answers incorrectly. On a controlled benchmark, a strong RAG-memory falls from 100% to 50% current-value accuracy as a fact is updated $2 \rightarrow 12$ times. We propose **Tenet**, which reframes memory as a **self-consistent belief state**—a compact *world model of the user*—rather than a document store. Tenet (i) distills raw turns into atomic, keyed facts; (ii) maintains a **bi-temporal** record so a changed fact *supersedes* its predecessor (retired to history, not overwritten); (iii) enforces **belief–evidence consistency** by retiring raw evidence that echoes a superseded belief; (iv) applies a **predictive-coding write policy**—surprise-gating—that stores only observations the model cannot already predict; and (v) closes the accuracy gap to raw retrieval with **belief-anchored evidence expansion**—spending spare context on query-relevant turns from the sessions the belief state already surfaced. Tenet holds 100% **current-value accuracy across all churn levels** (where a strong RAG-memory falls to 50%), matches strong RAG on retrieval recall (95–97.5%), and—with expansion—**matches its one-shot answer accuracy at equal-or-lower token budget** (57.5% vs 57.5%, gpt-4o reader) while retaining a high-efficiency point at **half the context** and the **best accuracy-per-token** of the systems we evaluate. Tenet thus traces an accuracy–efficiency *frontier* that meets RAG at its budget and beats it at every lower one. We release all code.

1 Introduction

An agent that talks to a user for months does not need a transcript; it needs a *model of the user* that stays true as the user changes. Yet the dominant memory design—retrieval-augmented generation over stored turns [1, 2]—treats memory as a document index: embed every turn, retrieve the top-k most similar, let the reader sort it out. This works for one-shot recall of a *static* fact. It fails, quietly, when facts **change**.

We formalize this as **knowledge churn**. A user who moves cities several times generates many "I moved to X" turns; all are similar to "where do I live?", so the top-k fills with *stale* versions. Once the number of updates exceeds k, the latest value may not be retrieved at all, and even when it is, the reader must infer recency from contradictory statements. Accuracy collapses (§4.2).

The root problem is abstraction. A document store has no notion that "I live in Boston" and "I live in Seattle" are the *same fact* with a *changed value*. We argue memory should be a **belief state**: a compact set of *current* beliefs, each with a temporal extent, updated by observation, kept internally consistent, and queryable across time—the stance world-model and predictive-coding accounts take toward perception [14], brought to agent memory.

Contributions.

1. We identify and name **knowledge churn**, a failure mode of retrieval memory, and give a controlled benchmark exhibiting it (RAG: 100%→50%; §4.2).
2. We present **Tenet**, a memory that is a self-consistent belief state: bi-temporal supersession, a **belief–evidence consistency** rule, and a **surprise-gated** (predictive-coding) write policy.
3. We evaluate on LongMemEval_S and controlled tests: Tenet is **churn-robust (**100%**)**, on par with RAG on recall (95%), **best on accuracy-per-token**, and—with belief-anchored evidence expansion—**at parity with strong RAG on one-shot accuracy at equal token budget**, closing a gap belief-only compression left open.
4. On the standardized **MemoryAgentBench FactConsolidation** benchmark, ingestion-time supersession with zero-LLM deterministic keys **exceeds the published single-hop SOTA at the gpt-4o-mini tier (86.5 vs 78.0 pooled) and ties the multi-hop tier (30.2)**—using only a local 7B backbone, where the benchmark's 22 systems score ≤60 / ≤7.

2 Related Work

Retrieval memory. Mem0 [1] distills salient facts at write time over a vector store; it attaches only a *creation* timestamp and *removed* its graph variant after finding it 3× slower / 2× tokens for a thin gain—evidence we weigh in choosing a light substrate. LongMemEval [2] is the standard long-horizon benchmark; its V2 adds a *latency-aware* metric, a field shift toward accuracy *per cost* that our per-token results target. **Temporal knowledge graphs.** Zep/Graphiti [3] maintain a bi-temporal graph with automatic invalidation—but pay heavy per-write extraction and require graph infrastructure; Tenet keeps the bi-temporal semantics without the graph. **The 2026 bi-temporal convergence.** Concurrent systems adopted bi-temporal supersession: MemStrata [7] applies deterministic (s, r, o) supersession over a bi-temporal ledger with no LLM on the read path—and shows *similarity-threshold* supersession leaks stale values where deterministic keying does not, independently corroborating our keyed design; Engram [8] pairs a bi-temporal graph with a hybrid facts-plus-chunks read path (converging on our dual-pool finding); TOKI [9] formalizes contradiction resolution as a bitemporal operator algebra; and [6] sets the current MemoryAgentBench FactConsolidation SOTA with deterministic post-retrieval aggregation. Tenet differs in *where* consistency is enforced—at ingestion and at recall (belief–evidence consistency)—and in coupling the belief state to surprise-gated writes and budget-bounded evidence expansion, evaluated on the knowledge-churn axis none of them report. **Memory in the Qwen ecosystem.** Alibaba's own 2026 memory line takes the opposite design point: QwenLong-L1.5 [11] reaches 76.4 on LongMemEval by RL-training a 30B reasoner that reads up to 128K tokens per query; AgeMem [12] RL-trains the memory policy itself (Qwen2.5-7B backbone); ActMem [13] (Alibaba-co-authored) reaches 75.6 with LLM-built causal graphs at ingestion. All three spend heavily—at training time, ingestion time, or read time. Tenet reaches 60.0 at the same reader tier with **zero-LLM ingestion and ∼2K read tokens per query** (∼79% of QwenLong-L1.5's score at under 2% of its read budget), and its structural wins survive on a 7B backbone. **OS-style / observational memory.** MemGPT/Letta [4] page memory between context and archival storage; observational schemes keep a stable summary. Both are largely append-oriented and do not treat supersession or forgetting as first-class. **What is missing.** No prior system combines bi-temporal supersession, *belief–evidence consistency*, *predictive-coding write-gating*, principled forgetting, and *belief-anchored evidence expansion* in a light, graph-free store—nor reports the knowledge-churn regime.

3 Method

Tenet stores two layers over one bi-temporal table: a **belief layer** of distilled, keyed facts, and an **evidence layer** of raw turns. Reads never call an LLM.

Distillation. Each turn is distilled into atomic facts, each with a stable semantic key $\kappa = \textit{subject}$: *attribute* (e.g. `user:residence`), a salience $s \in [0, 1]$, and an event time. The key makes supersession reliable: embedding similarity cannot separate a *restated* fact from a *value-changed* one (we measure a value change at cosine 0.99, indistinguishable from a paraphrase), but a shared key can.

Bi-temporal supersession. Every memory carries event time (`valid_at`, `invalid_at`) and transaction time (`created_at`, `expired_at`). Storing a fact at key κ whose value differs from the current one *supersedes* it: the old fact's `invalid_at`/`expired_at` are set; it leaves the current set but stays in history. Current recall filters `expired_at` $= \varnothing$; recall(as_of=t) reconstructs the belief set at any past t (time-travel).

Belief–evidence consistency (key rule). The evidence layer answers detail questions but is where stale values hide: "I moved to Boston" survives after the belief moves on. We retire from current recall any raw slice e close to a *superseded* belief:

$$\text{exclude } e \iff \max_{f \in \mathcal{F}_{\text{expired}}} \cos(e, f) \;\geq\; \tau_{\text{stale}}, \qquad \tau_{\text{stale}} = 0.80.$$

This single rule turns supersession into end-to-end correctness: 55% → 100% (§4.3).

Predictive-coding write policy. A world model stores *prediction error*. On write, a raw observation e is discarded if the store already predicts it:

$$\text{store } e \iff \max_{e' \in \text{store}} \cos(e, e') \;<\; g_{\text{surprise}}, \qquad g_{\text{surprise}} = 0.97.$$

This bounds the store and drops redundant repetition with no accuracy loss (§4.3).

Forgetting. Rank = relevance × decay, with $d = 2^{-\Delta t/h}\,(1 + \beta \log(1{+}\text{uses}))\,(0.6 + 0.8s)$, half-life $h = 14\,\text{d}$; a sweep archives low-d unpinned memories. Retrieval is a *dual pool* (beliefs for consistency, evidence for verbatim detail), each guaranteed a budget share.

Belief-anchored evidence expansion. Compressing a session into a few beliefs wins churn and efficiency but can drop the detail a multi-hop question needs, even when the right session *is* retrieved (recall 95–100%). The top-k result names both the belief state *and the sessions it came from* (each memory's `source`). Given spare budget B, Tenet fills it with query-relevant raw turns drawn *only from those already-surfaced sessions*—evidence anchored to the belief state, not the whole haystack—under the same stale-echo filter. Setting B to the baseline RAG budget makes expansion never spend more context than flat retrieval; the count m is a knob (m=0 → efficiency point; m large under B → parity point) that lets one system trace a frontier.

4 Experiments

Protocol. LongMemEval_S (500 questions, ∼115k-token histories). We use a `gpt-4o` **reader** (the same reader Mem0 and Zep report against), a local embedder (`bge-small`), and a cheap `gpt-4o-mini` distiller; the shipped system runs the same code on Qwen Cloud (`text-embedding-v4`, `qwen3.7-plus`) by a config flip. Numbers are compared only to baselines run under identical settings. Baselines: **RAG** (top-k raw turns) and **full-context** (entire history).

4.1 Recall, and the accuracy–efficiency frontier (n=40, gpt-4o reader)

System	mode	recall@10	QA acc	reader tok	**acc/1k tok**
full-context	—	—	65%*	∼124k	0.5*
RAG	top-k turns	95%	57.5%	2,101	27.4
Tenet	efficiency (m=0)	**97.5%**	52.5%	**1,067**	**49.2**
Tenet	parity (expansion)	**97.5%**	**57.5%**	2,083	27.6

Tenet is a *frontier*, not a point. At its **efficiency** point it answers at half the context (1,067 tok) for the best accuracy-per-token of any system (49.2, 1.6× RAG) at recall parity. Turning up belief-anchored expansion under RAG's own budget brings raw QA accuracy to **parity with a strong RAG (**57.5%=57.5%**) at fewer tokens**, closing the one-shot gap belief-only compression left open. Tenet thus meets RAG's accuracy at its budget and beats it at every lower one; the one category still behind is *multi-session* synthesis (42.9 vs 57.1, up from 28.6), where a question needs several evidence sessions but only some are surfaced. *Full-context, at a weaker reader, spends 100× the tokens for no gain over RAG. Reader-robust: on a cheaper `gpt-4o-mini` reader the parity point edges ahead (Tenet 60.0 vs RAG 55.0); per-token dominance holds across `gpt-4o-mini`, `gpt-4o`, and `claude-opus-4.8` (≈1.6–1.7×).

4.2 Knowledge churn (headline)

One fact updated N times amid distractors, $k = 6$, 12 principals/point:

updates N	2	4	6	8	10	12
RAG	100	100	100	67	58	**50**
Tenet	100	100	100	**100**	**100**	**100**

RAG degrades monotonically once $N > k$ (−50 pp); Tenet is flat at 100%. Supersession keeps exactly one current value regardless of churn—the property a *belief state* has and a document index cannot. **The curves are identical under a `gpt-4o-mini` and a `gpt-4o` reader**: the failure is structural (once $N > k$ the latest value is not reliably retrieved), so a stronger reader cannot rescue RAG.

4.3 Ablations

Belief–evidence consistency. Removing the rule drops Tenet to 55% current-value accuracy with a 45% stale-leak; adding it restores 100% / 0% leak—the single most important mechanism. **Surprise-gating** discards 15% of observations as redundant with no accuracy change, yielding a bounded store where RAG grows unboundedly.

4.4 Standardized conflict resolution: MemoryAgentBench FactConsolidation

On the ICLR 2026 conflict-resolution benchmark [5] (SubEM metric and official reader prompt verbatim; all 800 questions; Wilson 95% CIs), ingestion-time supersession with **fully deterministic, zero-LLM keys** and a deliberately weak **local 7B backbone** scores, pooled over four haystack lengths (6K–262K):

pooled	naive-RAG (same reader)	**Tenet**	published SOTA (mini / gpt-4o) [6]
FC-SH (n=400)	47.8	**86.5** [82.8, 89.5]	78.0 / 94.8
FC-MH (n=400)	4.5	**30.2** [26.0, 34.9]	30.2 / 51.5

Single-hop **exceeds the published gpt-4o-mini-tier SOTA (78.0; our CI excludes it) on a weaker backbone**; every system in the original benchmark table scores $\leq$60 (Zep 7, Mem0 18, MemGPT 28). Multi-hop exactly ties the mini-tier SOTA, where the original table's best is $\leq$7. Accuracy barely degrades with haystack length (SH 89 $\rightarrow$ 81 from 6K$\rightarrow$262K): the store is conflict-resolved at ingestion—the property assembly-time aggregation must re-derive at every read. Multi-hop still degrades with length (42 $\rightarrow$ 20); reported honestly.

To remove the backbone confound entirely, we reimplemented four published memory mechanisms as arms of the *same* harness (same 7B reader, embedder, SubEM, prompt; 6K+32K cells, n=200/axis): CAR [Freshness 2026] 87.5 SH / 33.0 MH, Mem0-style consolidation 81.0 / 12.0, HippoRAG-v2-style OpenIE+PPR graph 66.0 / 9.0, MemAgent-style overwrite memory 44.0 / 16.0 (n=25). **Tenet leads every arm on both axes (90.0 / 36.0 on the same cells)**—ingestion-time supersession matches or beats the best assembly-time aggregation while doing the consistency work once at write time.

4.5 Accurate retrieval at zero ingestion cost: MemoryAgentBench AR

On MAB's other core competency (∼2,000 questions, 197K–534K-token contexts; official per-benchmark metrics: SubEM for RULER-QA, the official LLM-judge for LongMemEval(S*), choice accuracy for EventQA; matched gpt-4o-mini reader), Tenet with **zero-LLM ingestion** (date-aware structured chunks + embeddings only) averages **59.3** — second only to HippoRAG-v2 [10] (65.1), which runs LLM OpenIE over every context token at ingestion, and 20+ points above Mem0 (32.6), Zep (37.5) and MemGPT. Per sub-benchmark: EventQA **70.7** (n=1,500; Wilson CI $[68.3, 72.9]$ excludes HippoRAG-v2's 67.6), RULER SH-QA 75.0 (parity with 76), LME(S*) 46.3 vs 50.7, and RULER MH-QA 45.0 vs 66 — the honest loss: Personalized-PageRank graph traversal is genuinely stronger at multi-hop chaining over narrative text. Together with §4.5, Tenet leads or ties the published field on two of MAB's four competencies while being the only system whose ingestion never calls an LLM.

4.6 Case study: web-agent trajectory memory (LongMemEval-V2)

Adapting Tenet's ingestion to LME-V2's web-agent trajectory haystacks [Wu 2026] (DOM states, actions; up to 115M tokens), three LLM-free changes—structure-aware chunking, query cleaning, and Qwen3-Embedding+BM25 hybrid retrieval (RRF)—lifted gold-evidence recall from a naive port's ∼12% to **59.7%** @48K budget (63.3% @72K; corpus ceiling ∼92%) at ∼0.25 s query latency. End-to-end accuracy is reader-gated, not retrieval-gated: 4- and 8-bit quantizations of the mandated Qwen3.5-9B reader both saturate at 45.1% [39.5, 50.8] ($P(\text{correct} \mid \text{gold}) \approx 0.6$ extraction ceiling); the substrate transfers, and the binding constraint is measurable and external to the memory.

5 Limitations

Multi-session synthesis is the one category where RAG still leads (42.9 vs 57.1). Belief-anchored expansion lifted it from 28.6 but does not close it: these questions need evidence from *several* sessions, and expansion only deepens the sessions the top-k already surfaced—if a required session is not among them, its detail is still missing. Session-diverse retrieval is the natural next step; elsewhere Tenet is at or above RAG. **The frontier is a knob, not free lunch**: parity accuracy costs RAG-equal tokens, and the 1.6× per-token win is at the efficiency point, which trades ∼5 pp of raw accuracy. **Evaluation**: n=40, off-Qwen (gpt-4o / gpt-4o-mini readers, local embedder), one seed; reader stochasticity is ≈±5–7 pp, so the one-shot result is reported as *parity*, not a win. The shipped system uses Qwen Cloud.

6 Conclusion

Treating agent memory as a **self-consistent belief state** rather than a document index makes it robust to the way real knowledge behaves: it changes. Tenet stays correct under knowledge churn where retrieval memory collapses, matches a strong RAG's one-shot accuracy at equal token budget while offering the best accuracy-per-token we tested, with a light graph-free substrate and no LLM in the read path—and gets time-travel and principled forgetting for free. We hope *knowledge churn* becomes a standard axis for evaluating agent memory.

References

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